

Patrick Amadeus Irawan

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RESEARCH INTERESTS

My research goal is to build **robust, self-improving multimodal agents** that perceive, reason, and act reliably across diverse environments. I develop **evaluation** and **post-training** methods for **vision-language and video models**, with a current focus on **interactive environments**. My prior work spans **VLM post-training** and **knowledge distillation**, multimodal reasoning and robustness, and multilingual and multicultural benchmarks.

Research Areas: multimodal reasoning and evaluation, multimodal post-training, interactive environments and world models, embodied AI

EDUCATION

Ph.D. in Natural Language Processing — MBZUAI	2025 - 2029
· Advisor: Alham Fikri Aji	
Global Study Program — University of California, Davis	2023
· computational neuroscience, tech management	
Bachelor in Computer Science — Institut Teknologi Bandung	2020 - 2024
· visual question answering; synthetic data generation; reasoning LLM	

PUBLICATIONS

* (Co-)first author † Second author

Ego2Act: Evaluating Goal-Directed Manipulation in Egocentric Video Generation*	Preprint 2026
<i>video generation, world model, evaluation, multi agent system, embodied agents</i>	
LinguDistill: Recovering Linguistic Ability in Vision Language Models via Selective Cross-Modal Distillation*	Preprint 2026
<i>VLM, SFT, post-training, knowledge distillation, architecture</i>	
Anthropogenic Regional Adaptation in Multimodal Vision-Language Model	AACL 2026
<i>VLM, SFT, post-training, data curation, evaluation, multilingual, multimodal</i>	
Vision Language Models are Confused Tourists*	CVPR 2026
<i>VLM, visual reasoning, robustness, hallucination, interpretability</i>	
A Massive-Scale Multilingual Multi-Cultural Multimodal RAG*	CVPR 2026
<i>RAG, multilingual, multimodal, evaluation, test-time scaling</i>	
Counting to Four is Still a Chore for VLMs*	CVPRW 2026
<i>VLM, numerical reasoning, interpretability, RL</i>	
Crosslingual Language Modeling Entropy as End-to-End Learnable Language Representations*	EMNLPW 2025
<i>LLM, multilingual, representation learning, unsupervised learning</i>	
WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural VQA on Global Cuisines*	NAACL 2025 (Best Theme Paper)
<i>VQA, multilingual, multicultural, benchmark, evaluation, data curation</i>	
Seeing Culture: A Benchmark for Visual Reasoning and Grounding†	EMNLP 2025
<i>visual grounding, evaluation, object detection, multicultural</i>	
Predicting Language Model Performance on Multilingual Tasks via Proxy Models	NAACL 2025
<i>multilingual, evaluation, efficient, regression, SFT</i>	
Towards Efficient and Robust VQA-NLE Data Generation with Large Vision-Language Models*	COLING 2025
<i>VQA, VLM, NLP, synthetic data generation, visual reasoning</i>	
SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages	EMNLP 2024
<i>multilingual, multicultural, video understanding, speech understanding, evaluation</i>	

WORK EXPERIENCE

Research Engineer	Feb 2025 – Sep 2025
<i>Singapore Management University</i>	
- Discovered up to a 30% performance degradation in frontier VLMs on a multicultural visual grounding & VQA benchmark, resulting in an EMNLP benchmark publication and an analysis-focused preprint. - Reduced data validation timeline by days for 50K+ spatial annotations with an automated VLM-based evaluation pipeline that automatically flagged poor polygon segmentation via foreground-color ratios. - Minimized model guessing shortcuts and ensured benchmark robustness by engineering a heuristic-based question-generation engine to synthesize hard adversarial distractors.	

Software Engineer

IT Bauschmiede

- Automated manual enterprise operations by integrating real-time scheduling, invoicing, and geolocation systems serving 3+ scaleup companies.
- Cut system latency by ~20% while maximizing uptime and scalability by architecting a microservices migration from a legacy monolithic MVC infrastructure.

2024–2025

Data Science, SWE, Research Internships

Ruangguru, Blibli, Supertype

- Scaled command-line autograder software to support 1k+ concurrent users with 99% uptime.
- Contributed to an 8% increase in absolute approved test-case volume within 6 months at major corporation scale.
- Optimized a topic extraction pipeline, resulting in a 2x efficiency gain and 20% classification improvement.
- Led the development of real-time AR and RAG multimodal systems for a high-stakes demo of 20k+ audiences.

2022–2024

ACHIEVEMENT & AWARDS

Columbia ML Summer School Scholarship	2026
OpenAI Micro Grant Award Grantee	2025
NAACL Best Theme Paper Award Winner	2025
BCG SEA Emeralds Case Competition Winner	2024
Indonesian International Student Mobility Awards @ UC Davis Awardee	2023
DOMO Higher Education Case Competition Finalist	2023
CSLeaders Full Bachelor Tuition Scholarship Awardee	2022
Gemastik XV Data Mining Division Finalist	2022

ADVISING & MENTORSHIP

Rava Maulana — <i>world models and evaluation</i>	now Software Engineer at Tiket
Iskandar Muda — <i>world models and evaluation</i>	now M.Sc. student at Bandung Institute of Technology
Qinrong Cui — <i>test-time adaptation for fine-grained visual understanding</i>	now M.Sc. student at MBZUAI
Wilfried Ariel — <i>multicultural object detection and evaluation</i>	now Ph.D. student at NTU

REFERENCES

Alham Fikri Aji — Assistant Professor, MBZUAI	alham.fikri@mbzuai.ac.ae
Genta Indra Winata — Principal Applied Researcher, Capital One	genta.winata@capitalone.com
Samuel Cahyawijaya — Member of Technical Staff, Cohere	samuel.cahyawijaya@gmail.com